

KITAMIESASHI, Japan

WMO: 474020

Lat: 44.9400N

Lon: 142.5850E

Elev: 27

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.6	10.6	0.7	5.7	9.3	3.5	6.6	11.8	29.4	29.6	26.0	28.5	6.4	240	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	81.0	69.4	77.6	68.0	74.5	66.9	72.3	78.2	70.4	75.2	68.5	72.9	12.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.3	111.6	76.2	68.4	104.6	73.9	66.7	98.2	72.0	36.1	78.2	34.4	75.4	32.9	72.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.9	20.5	17.5	DB	3.3	87.7	3.3	3.2	0.9	90.0	-1.0	91.9	-2.9	93.8	-5.3	96.1
			WB	2.6	74.9	3.3	2.4	0.2	76.6	-1.7	78.0	-3.5	79.4	-5.9	81.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	21.5	21.4	29.1	40.0	48.8	54.8	62.1	66.9	62.1	50.5	37.3	26.2	(6)
	DBStd	17.03	5.56	6.53	6.56	6.91	8.34	7.03	6.54	5.61	5.37	6.09	7.55	5.73	(7)
	HDD50	3992	885	800	647	312	128	26	0	0	0	68	388	738	(8)
	HDD65	8043	1350	1220	1112	750	507	318	138	43	120	450	832	1203	(9)
	CDD50	1622	0	0	0	12	89	169	376	525	363	83	5	0	(10)
	CDD65	197	0	0	0	0	3	10	49	103	32	0	0	0	(11)
	CDH74	785	0	0	0	0	35	58	208	396	87	1	0	0	(12)
	CDH80	139	0	0	0	0	5	9	41	76	8	0	0	0	(13)
(14) Wind	WSAvg	7.5	7.9	7.5	8.3	8.3	7.8	6.5	5.9	6.6	7.6	8.1	7.9	8.0	(14)
(15) Precipitation	PrecAvg	43.2	2.7	2.1	2.3	2.2	2.7	2.8	4.8	5.1	5.6	4.4	4.5	3.9	(15)
	PrecMax	58.6	5.3	4.6	4.5	4.9	5.5	7.7	11.7	13.0	14.5	6.6	8.8	5.5	(16)
	PrecMin	27.3	1.3	0.9	0.2	0.2	0.8	0.8	0.6	1.4	1.9	1.7	2.5	2.2	(17)
	PrecStd	6.9	1.0	0.9	1.0	1.1	1.2	1.4	2.8	3.0	2.8	1.5	1.7	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	43.2	50.2	67.8	79.2	80.4	84.8	86.1	80.4	69.8	60.3	45.4	(19)
		MCWB	33.0	37.1	41.0	51.8	58.4	62.7	72.7	73.3	68.3	59.5	51.5	41.1	(20)
	2%	DB	33.4	38.2	45.2	60.2	71.2	74.0	79.2	81.8	76.3	65.5	55.4	40.4	(21)
		MCWB	30.9	33.5	38.4	47.8	55.8	60.7	69.8	71.2	66.5	56.6	48.3	36.8	(22)
	5%	DB	31.8	34.4	41.5	54.8	66.1	68.9	75.5	78.4	73.2	62.6	51.8	37.3	(23)
		MCWB	29.7	30.8	36.2	44.7	53.7	59.3	68.6	69.8	65.0	55.5	46.4	34.3	(24)
	10%	DB	29.9	31.1	38.4	50.6	61.2	65.0	71.8	75.1	70.5	60.0	48.3	34.8	(25)
		MCWB	27.8	28.4	33.8	42.7	51.6	58.3	66.7	68.9	63.6	53.7	43.9	32.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.8	37.5	42.6	53.0	60.6	66.1	74.5	75.3	71.3	61.5	53.3	41.8	(27)
		MCDB	36.0	42.3	48.8	64.7	72.8	76.2	81.5	84.4	77.9	67.2	58.9	45.0	(28)
	2%	WB	31.7	33.9	39.2	49.0	57.3	62.9	71.6	73.1	69.0	58.6	49.5	37.5	(29)
		MCDB	33.1	37.6	44.4	58.6	68.4	70.9	77.7	79.4	73.7	63.3	54.1	39.9	(30)
	5%	WB	29.7	31.0	36.5	45.8	54.9	60.7	69.2	71.4	66.5	56.6	46.7	34.7	(31)
		MCDB	31.5	33.9	40.8	53.2	64.8	66.9	74.4	76.5	71.2	61.3	50.7	36.8	(32)
	10%	WB	27.9	28.7	34.2	43.4	52.3	58.7	66.8	69.7	64.6	54.6	44.2	32.4	(33)
		MCDB	29.8	31.1	37.8	50.0	59.9	64.1	71.3	73.6	69.3	59.1	48.3	34.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	11.4	10.9	12.2	12.4	10.3	8.9	9.3	12.7	13.3	10.3	9.1	(35)
		MCDBR	10.3	14.3	15.1	19.9	21.5	18.5	16.1	15.0	15.7	16.3	15.2	11.4	(36)
	5% WB	MCWBR	9.3	11.6	10.6	11.3	10.6	9.5	8.6	7.5	8.8	10.3	11.0	9.8	(37)
		MCWBR	9.8	13.3	14.4	18.5	20.1	15.7	14.4	12.9	13.2	14.3	13.8	11.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.336	0.381	0.443	0.508	0.524	0.504	0.481	0.453	0.404	0.406	0.387	0.356	(40)
		taud	2.032	1.861	1.807	1.820	1.880	2.030	2.162	2.259	2.338	2.168	2.114	2.035	(41)
	Ebn at Noon		231	240	243	239	241	247	251	254	258	234	212	207	(42)
		Edh at Noon	33	48	59	63	62	53	46	40	34	35	31	30	(43)
(44) All-Sky Solar Radiation	RadAvg		384	671	1053	1404	1543	1529	1426	1327	1151	783	416	308	(44)
	RadStd		42	88	94	121	133	169	184	96	102	50	45	26	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+2.09	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page